Midterm project: Block Bootstrap to estimate the downfall of an investment portfolio

Introduction

Stock investors commonly hold portfolios containing a combination of exchange-traded funds (ETFs) and individual stocks. Understanding the risks associated with these portfolios is essential for making informed investment decisions.

Traditional financial risk analysis often relies on measures such as historical volatility (eg. beta, which measures a stock's volatility or systematic risk compared to the broader market) or standard deviation. Although useful, these measures have important limitations. Financial returns can exhibit volatility clustering, extreme losses, and dependence across assets, particularly during market downturns.

Thus, the central question motivating this project is: **Given a portfolio's composition and historical market behavior, how severe could its future losses become?**

For the midterm part, I'm going to use three most common ETF to demonstrate my method: QQQ (tracks NASDAQ), VOO (tracks SP500), SCHD (a famous ETF by Charles Schwab that includes some best companies with high dividends).

Method

Historical prices and daily returns

The `quantmod` package retrieves adjusted ETF prices; `ggplot2` and `scales` support the plotting functions. Historical adjusted prices cover January 2015 through December 2025. For reproducible rendering, the report uses `data/prices.csv` when available, and otherwise downloads data from Yahoo Finance. The CSV should contain adjusted prices in columns `VOO`, `QQQ`, and `SCHD`, ordered by trading date.

The daily simple return for each ETF is $R_t = (P_t - P_{t-1})/P_{t-1}$. The aligned return matrix has trading days in rows and assets in columns.

Moving block bootstrap: `riskpath_sim()`

The documented functions are implemented separately in `R/riskpath.R` and imported using `source()`. `riskpath_sim()` samples consecutive blocks of historical returns with replacement, retaining short-term dependence within each block and contemporaneous dependence across ETFs. Its inputs are the return matrix, portfolio weights (nonnegative and summing to one), initial value, simulation horizon, number of trajectories, and block size. It returns a numeric matrix with `horizon + 1` rows and `n_sim` columns, including day zero. The simulation assumes constant weights rebalanced daily and ignores transaction costs.

For the example, VOO receives 50%, QQQ 30%, and SCHD 20% of a $10,000 portfolio. We simulate 5,000 trajectories over 150 trading days with 10-day blocks.

```r
set.seed(123)
paths <- riskpath_sim(
  returns = returns,
  weights = c(0.5, 0.3, 0.2),
  initial_value = 10000,
  horizon = 150,
  n_sim = 5000,
  block_size = 10
)
```

Risk summary: `riskpath_metrics()`

Terminal portfolio loss is $L = 1 - V_{terminal}/V_{initial}$; negative loss means a gain. `riskpath_metrics()` takes the simulated paths, a VaR confidence level (default 0.95), and a maximum-drawdown threshold (default 0.20). It returns **Value at Risk (VaR)**, the 95th percentile of terminal losses; **Expected Shortfall (ES)**, mean loss in the upper 5% tail; **mean maximum drawdown**, the average of each trajectory's largest decline from its running peak; and **probability of a drawdown at least 20%**. The last output retains the name `probability_20pct_decline` for compatibility, even if a different threshold is supplied.

```r
results <- riskpath_metrics(paths, confidence = 0.95, threshold = 0.20)
print(results)
```

```
$VaR
[1] 0.1198028

$ES
[1] 0.179385

$mean_max_decline
[1] 0.111975

$probability_20pct_decline
[1] 0.0876
```

Results and visualization

All plotting implementations also live in `R/riskpath.R`; each plotting function returns a `ggplot` object. The report calls them without duplicating their code.

Simulated portfolio paths

`plot_riskpath_paths()` displays all simulated trajectories and five straight reference segments connecting the initial investment to the 5th, 25th, 50th, 75th, and 95th percentiles of terminal value. The segments summarize terminal percentiles; they are not individual simulated paths.

Value at Risk

`plot_riskpath_var()` displays the distribution of terminal losses, with a dashed red line at the 95% VaR and an annotation showing how many simulations exceed that cutoff.

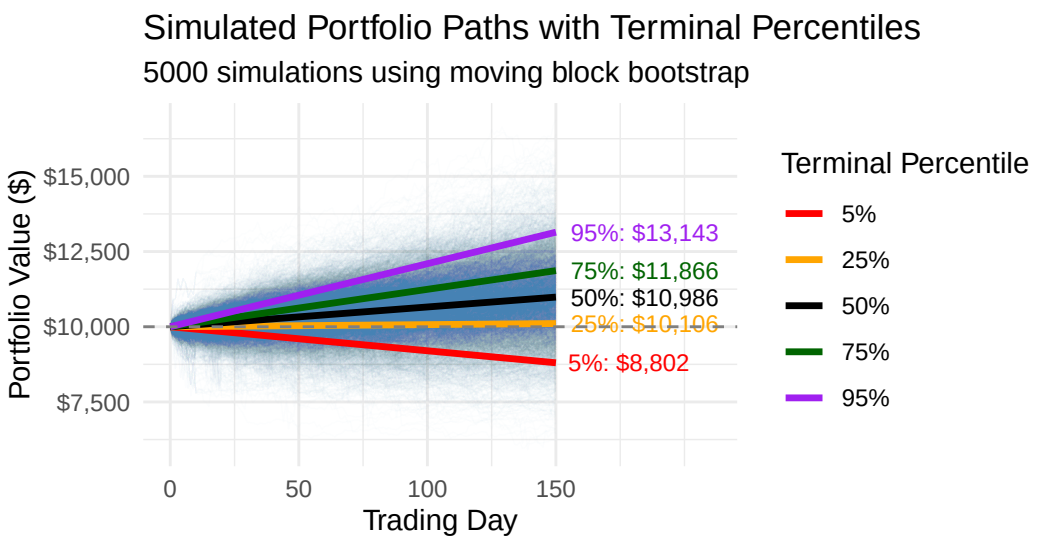


Figure 1: Simulated portfolio trajectories and terminal-value percentiles

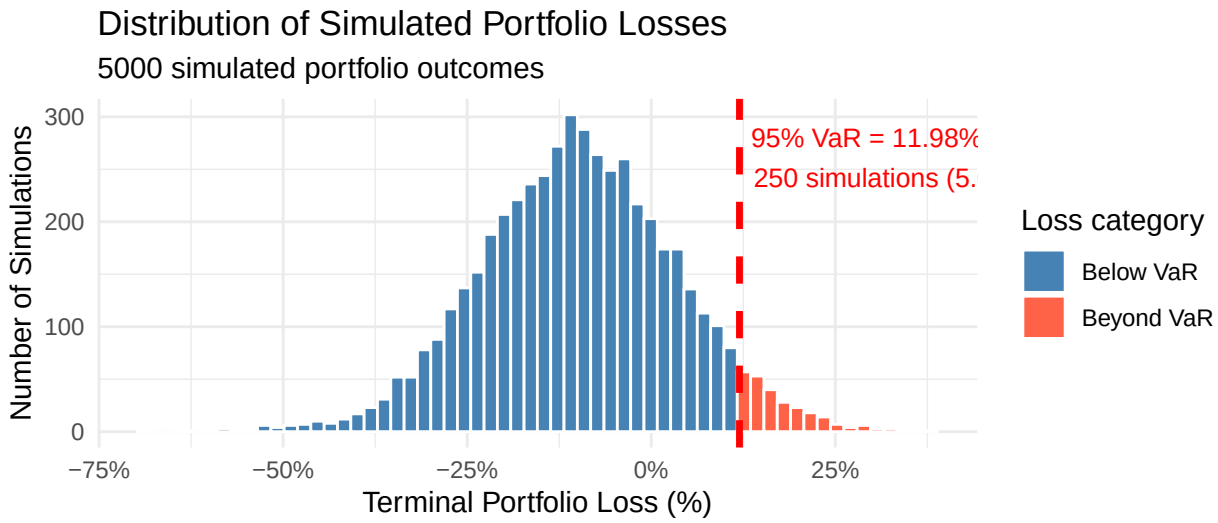


Figure 2: Distribution of terminal losses and 95% VaR

Maximum drawdown

`plot_riskpath_drawdown()` displays each path's maximum peak-to-trough decline. The black dashed line marks mean maximum drawdown and the red dashed line marks the 20% threshold.

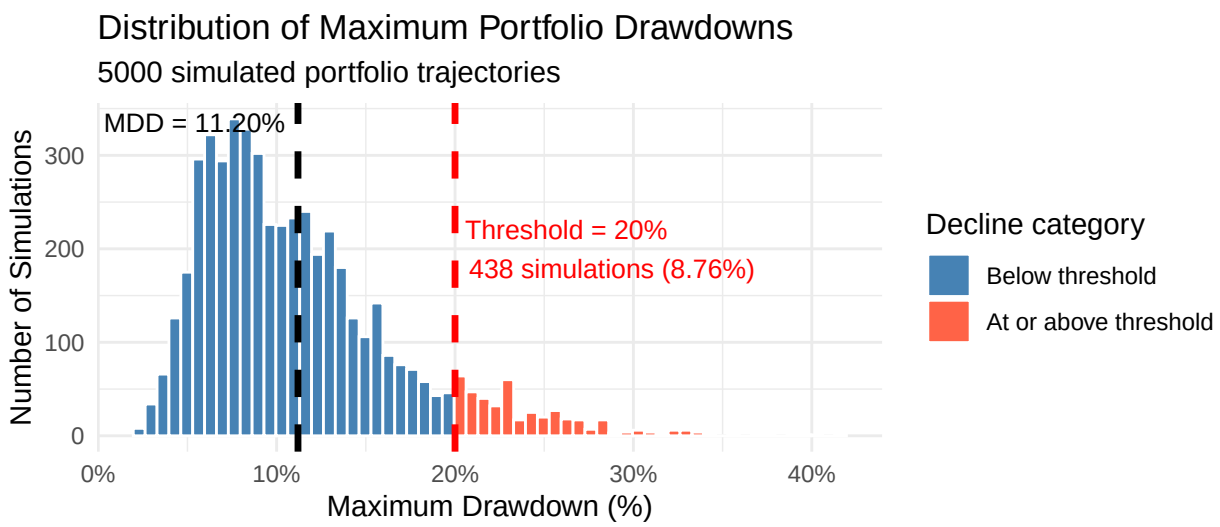


Figure 3: Distribution of maximum drawdowns with a 20% threshold

Interpretation and limitations

The bootstrap produces a distribution of possible portfolio outcomes conditional on historical ETF returns, the block length, and the fixed allocation. It preserves relationships within sampled 10-day blocks, but cannot guarantee that future market conditions resemble the past. The estimates do not account for transaction costs, taxes, or changes in portfolio allocation. They are scenario-based risk summaries rather than guaranteed forecasts.

AI Disclosure

I developed the idea of using a moving block bootstrap for portfolio simulation and described the intended workflow to AI, which helped generate the R implementation. I reviewed the code step by step, including unfamiliar functions such as `stopifnot()`, `unlist()`, and `cumprod()`, and checked that it followed my intended algorithm. I also used AI to help implement the visualizations from my proposed designs. The functions and plots are kept in the documented source file for inspection and reproducibility.